Analytics & Signal Validation

Sector-Wise Market Intelligence & Signal Dashboard · Oct 13, 2026 · @Adya

1. Introduction

Historical prices match the source and the computed signals are correct, but the accuracy check did not cover the newest rows, where a pipeline timing bug had stored intraday prices for 13 Oct 2026. With that fixed, the analysis found clear differences in sector risk, return and valuation. This section covers the work of Person B (Analytics & Signal Validation): checking that the database can be trusted, then using it to answer analytical questions.

All analysis uses the project database (PostgreSQL 17 on Neon), accessed through the read-only `powerbi_reader` login. It covers 25 NSE large-cap stocks in 5 sectors (IT, Banking, Auto, Pharma, FMCG) with daily prices from January 2015 to 13 October 2026, loaded from Yahoo Finance and refreshed every weekday at 5 PM IST. Queries were run in SQL, and checks and charts were built in a spreadsheet (Apple Numbers, exported to Excel).

2. Methodology

Each task used one SQL query (all saved in `person_b_queries.sql`) and one sheet of the analysis workbook.

#	Task	Method	Data used
1	Data accuracy	1 random stock per sector × 5 random trading days (25 prices); stored `close` compared with Yahoo Finance; match if difference < 0.5%	`daily_prices`, `companies`, `sectors`
2	Signal validation	20-day momentum, SMA-50 and RSI-14 recomputed independently with spreadsheet formulas for ICICI Bank and Maruti over the last 100 trading days	`daily_prices`, `signals`

#	Task	Method	Data used
3a	Strongest sector per year	Daily equal-weighted sector returns compounded into calendar-year returns, then ranked within each year	`v_sector_daily`
3b	Volatility	Annualised volatility = standard deviation of daily sector returns × √252	`v_sector_daily`
3c	Co-movement	Pearson correlation of daily returns for every pair of sectors (5×5 matrix)	`v_sector_daily`
4	Anomaly case studies	20 largest flagged anomalies ranked by size of move; 5 picked from different dates and causes; news found for each	`signals` (`is_anomaly`)
5	Backtest	Each month, hold the sector with the highest relative strength on the previous month-end; compared with equal-weighting all 5 sectors; growth of ₹100	`v_sector_strength` , `v_sector_daily`
6	Valuation	Each stock's P/E and P/B compared with its own sector median; Cheap if P/E < 80% of median, Expensive if > 120%	`fundamentals` (latest snapshot)
7	New signal	52-week high breakout logic designed; implemented in the database by the database engineer and verified	`daily_prices` , `signals`

The signal formulas used in Task 2 were:

- **20-day momentum** = today's adjusted close ÷ adjusted close 20 trading days earlier − 1
- **SMA-50** = average of the last 50 adjusted closes, including today
- **RSI-14** = 100 − 100 ÷ (1 + average gain ÷ average loss), using simple averages of the last 14 daily changes

All returns use adjusted close (`adj_close`), so dividends, splits and bonus issues do not appear as false price moves.

3. Findings: data accuracy and signal validation

All 25 sampled closing prices matched Yahoo Finance within ₹0.01. The sample covered ICICI Bank, ITC, Maruti Suzuki, Sun Pharma and Wipro on 30 Jun 2017, 13 Jul 2018, 24 Feb 2021, 28 Apr 2025 and 5 Jun 2026. The only differences (up to ₹0.005) came from Yahoo's web page rounding to 2 decimals while the database stores 4.

Two stocks needed explanation rather than correction. Wipro's older prices are lower than the prices that actually traded, because Yahoo adjusts history for its 2019 and 2024 bonus issues. ITC's pre-2025 prices are adjusted for the ITC Hotels demerger. Exchange (NSE) records show the unadjusted traded prices, so they differ from the database on those dates by design.

The random sample missed the newest data, and that is where an error turned up. The latest sampled date was 5 Jun 2026. A follow-up check found Maruti's 13 Oct 2026 close stored as ₹11,320 against NSE's official ₹11,245. The cause was a pipeline run at 09:39 IST, six hours before market close, which saved intraday prices for 24 of the 25 stocks; 16 of them were more than 0.5% off the official close.

The pipeline now refuses a same-day bar before 16:30 IST and re-checks the last 5 trading sessions on every run, and the 13 Oct rows were replaced with official closes. The lesson for the check itself: always include the most recent rows, not only random historical dates.

All three signals matched an independent recalculation for both stocks tested.

Signal	ICICI Bank: largest difference	Maruti Suzuki: largest difference	Tolerance	Result
20-day momentum	0.0000005	0.0000005	< 0.0001	Pass
SMA-50 (₹)	0.00005	0.00005	< 0.01	Pass
RSI-14 (points)	0.005	0.005	< 0.1	Pass

The remaining differences come from rounding in storage (RSI is stored to 2 decimals, SMA to 4). The check also confirmed that signals are computed on adjusted prices: recomputing on raw `close` gave SMA-50 errors of up to ₹11 and RSI errors of up to 4.7 points for ICICI Bank, because of its ₹12 dividend on 3 Aug 2026.

4. Findings: sector analysis

4.1 Strongest sector each year

Auto was the strongest sector most often (5 of 12 years), but leadership rotated almost every year. Auto was also the weakest sector in 3 years, so it is high-reward and high-risk.

Year	Strongest	Return	Weakest	Return
2026 (to 13 Oct)	Pharma	+5.4%	IT	−22.0%
2025	Auto	+37.7%	IT	−10.6%
2024	Pharma	+46.6%	FMCG	−7.1%
2023	Auto	+48.8%	Banking	+10.4%
2022	Auto	+25.2%	IT	−27.2%
2021	IT	+61.3%	Auto	+0.3%
2020	Pharma	+67.7%	Banking	+1.3%
2019	Banking	+29.1%	Auto	−6.9%
2018	FMCG	+30.3%	Auto	−13.3%
2017	Auto	+41.7%	Pharma	−5.4%
2016	Auto	+13.6%	Pharma	−15.7%
2015	FMCG	+16.7%	Banking	−7.0%

2020 shows the COVID pattern clearly: Pharma (+68%) and IT (+57%) surged while Banking barely moved. IT swung the most between years, from +61% in 2021 to −27% in 2022.

4.2 Most and least volatile sector

Banking is the most volatile sector (22.3% a year) and FMCG the most stable (16.2%).

Sector	Sector volatility (annualised)	Average stock volatility	Worst day	Best day
Banking	22.3%	25.7%	−17.0%	+10.9%
Auto	21.4%	26.2%	−12.6%	+11.0%
IT	21.2%	24.9%	−8.8%	+8.8%
Pharma	20.0%	26.1%	−7.3%	+10.0%
FMCG	16.2%	21.7%	−10.7%	+10.1%

Banking's worst day (−17%) was the COVID crash of 23 Mar 2020. FMCG is "defensive" because demand for food and household goods holds up in any economy. Every sector is calmer than its average stock (for example, Pharma 20.0% vs 26.1%), which shows diversification at work: holding 5 stocks together cancels out part of each one's swings.

4.3 How closely sectors move together

All sector pairs move in the same direction, but only moderately (correlation 0.31 to 0.60).

	Auto	Banking	FMCG	IT	Pharma
Auto	1.000	0.596	0.498	0.340	0.368
Banking	0.596	1.000	0.439	0.335	0.329
FMCG	0.498	0.439	1.000	0.349	0.362
IT	0.340	0.335	0.349	1.000	0.313
Pharma	0.368	0.329	0.362	0.313	1.000

Auto and Banking are the most closely linked (0.60): both depend on the domestic economy and interest rates. IT is the most independent (0.31–0.35 with every other sector) because its revenue comes mainly from US and European clients. Because no pair is close to 1, spreading money across sectors genuinely reduces risk.

5. Findings: anomaly case studies

Each of the 5 largest distinct anomalies traces to a real, identifiable event, which confirms the anomaly flag picks up genuine market news. They cover five different kinds of cause.

Date	Stock	Move	Cause	Category
23 Mar 2020	Axis Bank	−27.9%	COVID-19 crash	Market-wide shock
25 Oct 2017	State Bank of India	+27.7%	PSU bank recapitalisation plan	Government policy
21 Mar 2017	Divi's Laboratories	−19.8%	USFDA import alert on Vizag Unit-II	Regulatory action
22 Oct 2019	Infosys	−16.2%	Whistleblower complaint against CEO/CFO	Corporate governance
15 Jul 2020	Wipro	+16.8%	Q1 FY21 margins beat estimates	Quarterly results

Axis Bank, 23 Mar 2020 (−27.9%). Indian markets had their worst day as the country moved towards a nationwide lockdown, and banks fell hardest on fears of loan defaults. ICICI Bank, Maruti and Tech Mahindra crashed the same day, so this was a market-wide event rather than a company-specific one.

State Bank of India, 25 Oct 2017 (+27.7%). The government announced a plan of about ₹2.11 lakh crore to recapitalise public-sector banks the previous evening (Business Standard). PSU bank stocks surged and SBI led the Sensex and Nifty to record closes; its volume z-score of 64 is the highest in the anomaly list.

Divi's Laboratories, 21 Mar 2017 (−19.8%). The US FDA issued an import alert on the company's Visakhapatnam plant, blocking its products from the US market (Business Standard). As the US is a key market for Indian pharma, the stock fell about 20% in one day.

Infosys, 22 Oct 2019 (−16.2%). Anonymous whistleblowers accused the CEO and CFO of unethical practices to inflate short-term profits (Business Standard). It was the stock's worst fall in years, and the company ordered an independent investigation.

Wipro, 15 Jul 2020 (+16.8%). Wipro's Q1 FY21 operating margins came in well above analysts' expectations despite the COVID slowdown (Business Standard), and the stock jumped on very high volume (z-score 31.5).

One weakness of the detection method showed up: the COVID-day anomalies were flagged on price only, with volume z-scores between −0.4 and 2.2. Volume had already been very high for weeks, so that day did not look unusual against the previous 20 days. A rolling z-score can miss volume anomalies when the whole market is already in turmoil.

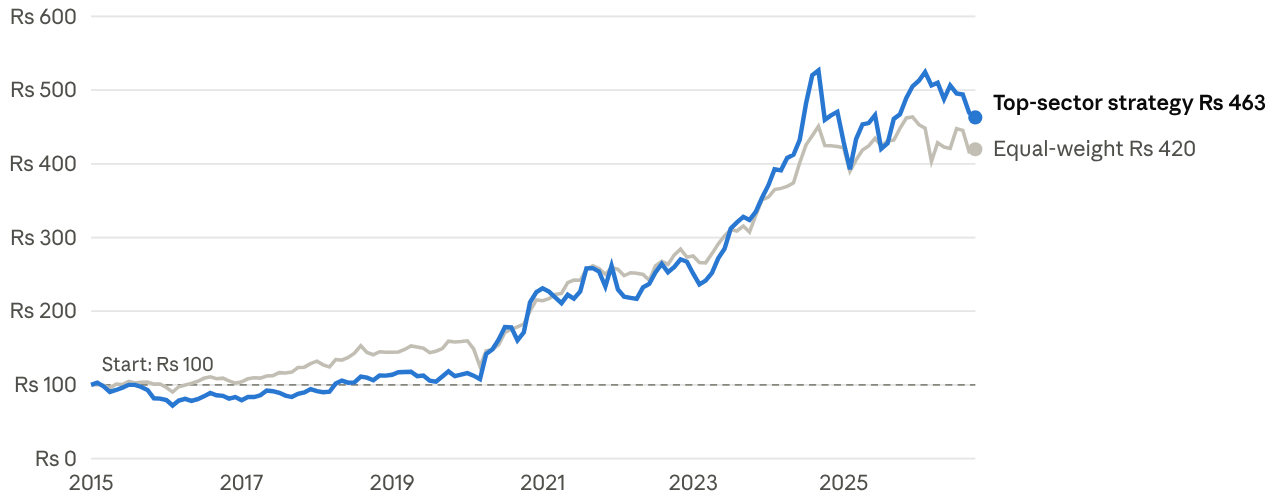
6. Findings: backtest and valuation

6.1 Does buying the top-ranked sector beat equal-weight?

Yes, but only slightly, and with much more risk. ₹100 grew to ₹463 by holding the top-ranked sector each month, against ₹420 by splitting money equally across all 5 sectors.

Top-sector strategy ended ahead, but trailed equal-weight until mid-2023

Value of Rs 100 invested in Feb 2015, monthly, no trading costs



Task 5 backtest query on v_sector_strength and v_sector_daily · Feb 2015 – Oct 2026, 141 months

Measure (Feb 2015 – Oct 2026)	Top-sector strategy	Equal-weight
Yearly growth (CAGR)	13.9%	13.0%
Volatility (annualised)	21.2%	14.2%
Worst fall from a peak	−29.9%	−21.7%

Measure (Feb 2015 – Oct 2026)	Top-sector strategy	Equal-weight
Best month	+31.5%	+16.9%
Worst month	−12.7%	−15.9%
Months the strategy did better	51%	—

Return per unit of risk was better for equal-weight (0.92 vs 0.66). The strategy beat equal-weight in only 51% of months; its lead comes from a few large winning months, such as holding Pharma in April 2020 (+31.5%). It also fell to ₹72 in early 2016 while equal-weight stayed near ₹100. Sector momentum added about 1 percentage point a year over 11 years, but not consistently, and frequent switching would add trading costs and taxes that this test ignores.

The last backtest month, October 2026, is only 13 days long (1–13 Oct), so the final values rest on a partial month and depend heavily on the 13 Oct closes. In that month the strategy (holding Pharma) returned −1.38% against +0.80% for equal-weight; with the stale 13 Oct prices the figures were −1.71% and −0.14%, and the final values ₹461 and ₹416.

6.2 Which stocks look cheap or expensive?

Valuations differ more between sectors than within them: the median P/E is 41.4 for FMCG but 15.5 for Banking, so each stock is judged against its own sector.

Sector	Median P/E	Cheap (P/E < 80% of median)	Expensive (P/E > 120% of median)
Auto	23.3	Mahindra & Mahindra (16.7), Hero MotoCorp (18.0)	Eicher Motors (33.5)
Banking	15.5	State Bank of India (10.3)	Kotak Mahindra Bank (21.6)
FMCG	41.4	ITC (16.1)	Nestle India (67.4)
IT	15.1	none	Tech Mahindra (26.2), HCL Technologies (18.9)
Pharma	31.2	Lupin (16.2)	Divi's Laboratories (84.9)

Fundamentals are the snapshot of 10 Oct 2026. Divi's is the most expensive stock (172% above its sector median), reflecting the premium the market pays for its expected growth. SBI trades at a deep discount to private banks, typical of PSU banks because of

government ownership and higher bad-loan risk. ITC's low P/E should be read with care, because its trailing earnings may include a one-time gain from the ITC Hotels demerger; its 6% dividend yield is genuinely high. A low P/E is a screening signal, not a buy recommendation: it can reflect real problems, just as a high P/E can be justified by growth.

7. New signal: 52-week high breakout (implemented)

A stock closing above its highest price of the previous 52 weeks is flagged as a breakout, and as a strong breakout when volume is also unusually high. It is now live in the `signals` table.

- **high_52w** = highest adjusted close over the previous 252 trading days, excluding today
- **breakout_52w** = TRUE when today's adjusted close is above high_52w
- **strong_breakout** = breakout_52w AND volume z-score above 2
- **pct_from_52w_high** = how far today's price is below its 52-week high, in %

The first 252 trading days of each stock have no value. Daily breakout counts per sector (breakout_count, strong_breakout_count) are in `v_sector_daily` for the dashboard.

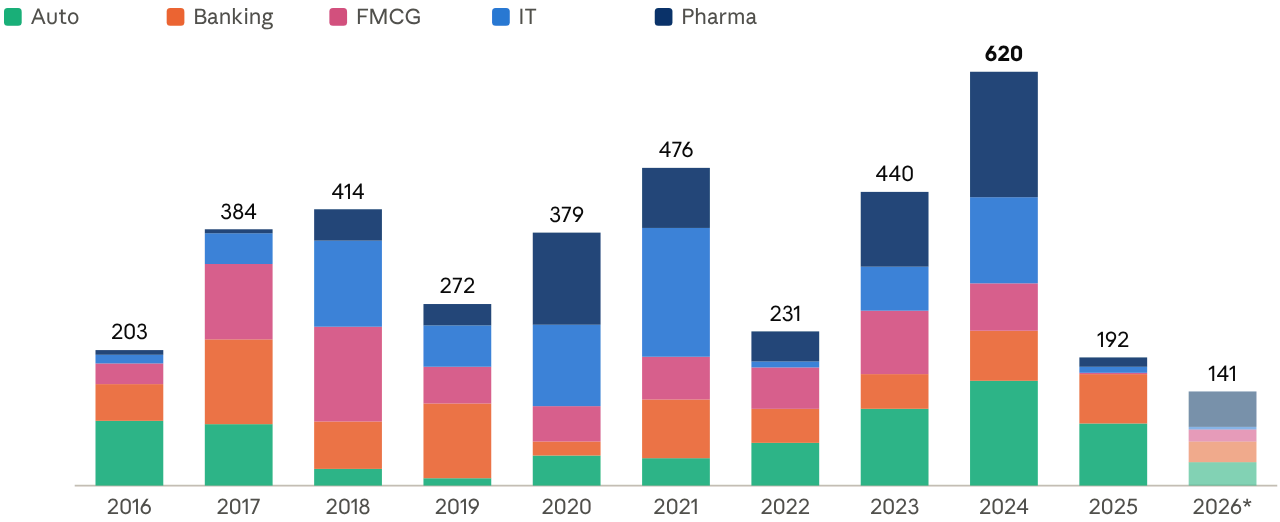
The signal complements the existing ones: momentum measures speed, RSI measures whether a stock is overbought, and a breakout shows a stock moving into new price territory, which often marks the start of a sustained uptrend.

What the breakout signal found

Breakouts track the sector leadership found in section 4: they cluster in a sector's strong years and almost vanish in its weak ones.

Breakouts peaked at 620 stock-days in 2024, then dried up in 2025

Stock-days closing above the prior 52-week high, by sector; 2015 has none by construction



* 2026 runs to 13 Oct only (lighter bar).

signals.breakout_52w via v_sector_daily.breakout_count · production database, 2016 – 13 Oct 2026

IT had 193 breakout stock-days in its +61% year (2021) but only 9 in 2022, when it fell 27%, and 4 so far in 2026. Pharma led 2020 (138) and 2024 (188), matching its best calendar returns. Breakouts fell from 620 in 2024 to 192 in 2025, with FMCG at 2, a sign the broad rally had faded. The most recent breakouts were Divi's Laboratories (23 Sep 2026) and Bajaj Auto (1 Sep 2026, on heavy volume); no stock broke out on 13 Oct. Of 3,752 breakout stock-days since 2016, 635 (17%) were strong breakouts on unusually heavy volume.

The signal was checked independently: 10 SQL validation checks pass, and a separate recomputation for ICICI Bank, Maruti and TCS matched all four columns with no mismatches.

8. Charts and tables to include

All are in the analysis workbook (`PersonB_Analysis.xlsx`), one sheet per task.

Figure	What it shows	Workbook sheet
Data accuracy table	25 database vs Yahoo prices, all "Match"	1_DataCheck
Signal check tables	Recomputed vs stored momentum, SMA-50, RSI-14 with pass/fail summary	2_SignalCheck, 2b_Maruti
Yearly sector returns	Strongest and weakest sector each year, 2015–2026	3a_Yearly

Figure	What it shows	Workbook sheet
Volatility table	Annualised risk, worst and best day by sector	3b_Volatility
Correlation heatmap	5×5 matrix, coloured by strength	3c_Correlation
Anomaly case table	5 events with date, move, cause and category	4_Anomalies
Growth of ₹100 line chart	Strategy vs equal-weight (shown in section 6.1)	5_Backtest
Valuation table	P/E vs sector median, colour-coded Cheap / Fair / Expensive	6_Valuation

All SQL used is in `person_b_queries.sql`.

9. Limitations

- **Daily data, not real-time.** Prices are end-of-day, updated at 5 PM IST on weekdays. The system cannot show intraday moves, and the latest row can briefly differ from other sources (on 13 Oct 2026 a run before market close briefly stored intraday prices as closes; see section 3).
- **Fundamentals are current values only.** The fundamentals table stores recent weekly snapshots, not history, so P/E cannot be compared over time and the backtest cannot use valuation.
- **Survivorship bias.** The 25 stocks are today's large-caps. Companies that shrank or left the index since 2015 are not included, which makes historical sector returns look better than an investor would actually have earned.
- **Simple RSI.** RSI-14 uses simple averages rather than the Wilder smoothing used by most trading platforms, so its values will differ from those shown on sites like NSE or TradingView.
- **Small sectors.** Each sector has only 5 stocks, so one stock's large move (such as Axis Bank on 23 Mar 2020) shifts the whole sector average.
- **Equal-weighted sectors.** Each stock counts equally regardless of size, unlike official indices such as the Nifty sector indices, which weight by market value.
- **Backtest simplifications.** No trading costs, taxes or slippage; one rule tested over one period, so the result may not repeat.

- **Adjusted prices.** Yahoo revises adjusted prices when new dividends are paid, so stored adjusted values for old dates can change slightly over time.

10. Conclusion

Historical prices match the source and all three signals recompute correctly on adjusted prices; the one data error found, intraday prices saved as the 13 Oct 2026 close, sat in the newest rows, which the original random check did not sample. Using it, the analysis found that sector leadership rotates every year, Banking carries the most risk and FMCG the least, IT moves most independently of the rest, and the largest anomalies line up with real news events. A simple top-sector momentum strategy beat equal-weighting by about 1 percentage point a year, but with roughly 50% more volatility, so equal-weighting gives the better balance of risk and return. A 52-week high breakout signal has now been implemented and is live on the dashboard; it flagged 3,752 breakout stock-days, 635 (17%) of them strong breakouts.

Sources

- Divi's Labs tanks 20% after USFDA issues import alert on Vizag unit – Business Standard
- Indian banking shares surge on government's $32.4 billion recapitalisation plan – Business Standard
- Infosys tumbles 16% as whistleblowers accuse of unethical practices – Business Standard
- Wipro surges 15% as Q1 operational performance beats Street estimate – Business Standard
- Yahoo Finance historical prices (data accuracy check)